19 A point source emits 50.0 W of sound energy in all directions. A small microphone of area 0.85 cm^2 detects the sound at 4.0 m from the source.

What is the power received by the microphone?

A 1.6×10^{-5} W

B 2.1×10^{-5} W

C 2.1×10^{-1} W

D 2.5×10^{-1} W

Intensity = power emitted /area spread = $50 / 4\pi r^2$

Power received = intensity $\times$ microphone area = 2.1×10^{-5} W

Answer: B

20 The diagram shows a standing wave on a string. The standing wave has three nodes N_1, N_2 and